

Development of the bile ducts: Essentials for the clinical hepatologist

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Summary

Several cholangiopathies result from a perturbation of developmental processes. Most of these cholangiopathies are characterised by the persistence of biliary structures with foetal configuration. Developmental processes are also relevant in acquired liver diseases, as liver repair mechanisms exploit a range of autocrine and paracrine signals transiently expressed in embryonic life. We briefly review the ontogenesis of the intra- and extrahepatic biliary tree, highlighting the morphogens, growth factors, and transcription factors that regulate biliary development, and the relationships between developing bile ducts and other branching biliary structures. Then, we discuss the ontogenetic mechanisms involved in liver repair, and how these mechanisms are recapitulated in ductular reaction, a common reparative response to many forms of biliary and hepatocellular damage. Finally, we discuss the pathogenic aspects of the most important primary cholangiopathies related to altered biliary development, i.e. polycystic and fibropolycystic liver diseases, Alagille syndrome.

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Introduction

Development of the biliary system is a unique process that has been thoroughly reviewed in several recent papers [1,2]. Here, we will focus on those concepts of biliary development that are “essential” to understand congenital and acquired cholangiopathies.

Cholangiopathies are an heterogeneous group of liver diseases, caused by congenital, immune-mediated, toxic, infectious or idiopathic insults to the biliary tree [3,4]. In addition to being responsible for significant morbidity and mortality, cholangiopathies account for the majority of liver transplants in paediatrics and a significant percentage of liver transplants in young adults. Many cholangiopathies are congenital, resulting initially from an altered development of the biliary tree, eventually accompanied by necro-inflammatory processes [5,6]. For the clinical hepatologist, this means a good working knowledge of the mechanisms of liver development is necessary for the care of these patients.

We will briefly review the general aspects of bile duct development and morphogenesis and the main molecular mechanisms involved in bile duct ontogenesis. Then we will highlight the role of these mechanisms in liver repair. Lastly, we will discuss the cholangiopathies related to altered development, with a special emphasis on those caused by a single genetic defect.

General aspects of bile duct morphogenesis during liver development

The liver develops as a tissue bud deriving from a diverticulum of the ventral foregut endoderm, which extends into the *septum transversum*, a structure located between the pericardial and peritoneal cavities. The ventral foregut endoderm develops two protrusions: the cranial part leads to the formation of the intrahepatic bile ducts, while the caudal part generates the extrahepatic biliary tree [1,7,8].

Intrahepatic biliary tree

The development of the intrahepatic biliary epithelium begins around the 8th week of gestation (GW), and proceeds centrifugally from the hilum to the periphery of the liver following the portal vein system [5,6,9]. At birth, the intrahepatic biliary epithelium is still immature, and its maturation is completed during the first years of life [9]. The sequence of events leading to the development of the ductal plate and the intrahepatic bile ducts

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Abbreviations: GW, week of gestation; Sox9, SRY-related HMG box transcription factor 9; K, cytokeratin; HNF, hepatocyte nuclear factor; TGF, transforming growth factor; TβRII, transforming growth factor receptor type II; PCP, planar cell polarity; IL, interleukin; Hh, Hedgehog; Shh, sonic Hedgehog; DPM, ductal plate malformations; VEGF, vascular endothelial growth factor; PBP, peribiliary plexus; RBP-JK, recombination signal binding protein for immunoglobulin kappa J; AGS, Alagille syndrome; Dvl, Dishevelled; IGF1, insulin-like growth factor-1; CTGF, connective tissue growth factor; SDF-1, stromal cell-derived factor 1; TNFα, tumour necrosis factor-α; NCAM, neural cell adhesion molecule; HPC, hepatic progenitor cells; IHBC, intermediate hepato-biliary cells; RDC, reactive ductular cells; MKS, Meckel syndrome; ARPKD, autosomal recessive polycystic kidney disease; CHF, congenital hepatic fibrosis; CD, Caroli's disease; FPC, fibrocystin; ADPKD, autosomal dominant polycystic kidney disease; PLD, polycystic liver disease; PC, polycystin; PKA, protein kinase A; ER, endoplasmic reticulum; CCA, cholangiocarcinoma; MCP-1, monocyte chemoattractant protein-1; CINC, cytokine-induced neutrophil chemoattractant; ET-1, endothelin-1.



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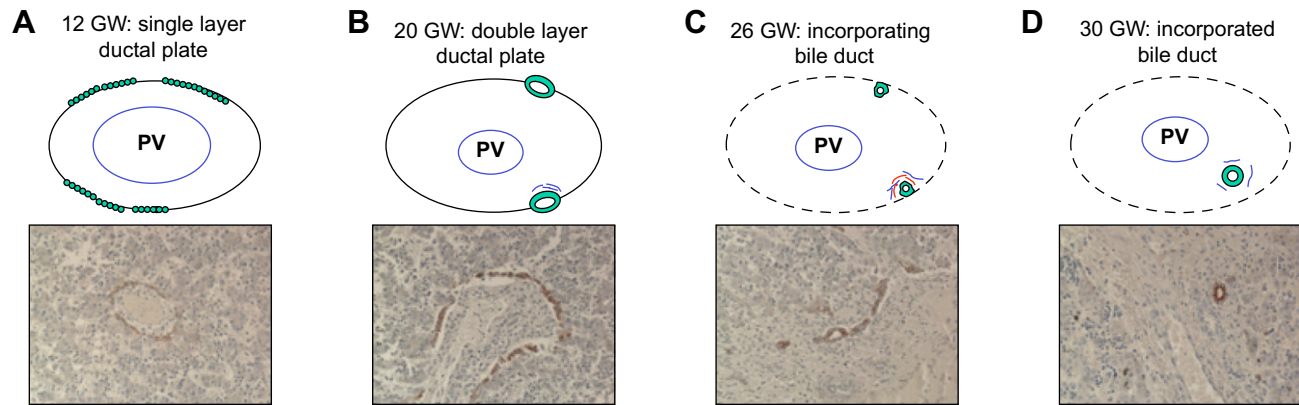


Fig. 1. Embryological stages of intrahepatic bile duct development. The development of intrahepatic bile ducts starts when the periportal hepatoblasts, in close contact with the portal mesenchyme surrounding a portal vein branch, begin to organise into a *single layered* sheath of small flat epithelial cells, called *ductal plate* ("ductal plate stage", A). In the following weeks, discrete portions of the ductal plates are duplicated by a second layer of cells (*double layered ductal plate*, B), which then dilate to form tubular structures in the process of being incorporated into the mesenchyme of the nascent portal space (*incorporating bile duct*, "migratory stage", C). Once incorporated into the portal space, the immature tubules are remodelled into individualised bile ducts (*incorporated bile duct*, "bile duct stage", D).

are shown in Fig. 1 [10–12]. Whether or not segments of the ductal plate, that are not incorporated into the nascent bile ducts, are gradually deleted by apoptosis is a matter of controversy. Recent data from Lemaigre's group [13] suggest that ductal plate remodelling does not occur by proliferation and apoptosis. Rather, portions of the ductal plate appear to differentiate into periportal hepatocytes and adult hepatic progenitor cells [13]. The role of ductal plate cells as potential stem cells has been recently addressed by Furuyama *et al.* [14]. A crucial event in the development of the intrahepatic biliary system, as well as in liver repair, is tubulogenesis. Biliary tubule formation depends upon a unique process of *transient asymmetry*. Careful studies performed in mouse embryos have shown that nascent tubules are formed by ductal plate cells resembling cholangiocytes (positive for the SRY-related HMG box transcription factor 9 [Sox9] and cytokeratin-19 [K19]) on the side facing the portal tract, and by ductal plate cells resembling hepatoblasts (positive for the hepatocyte nuclear factor 4 [HNF4] and the transforming growth factor receptor type II [T β RII]) on the side facing the parenchyma. The "portal" layer shows higher levels of E-cadherin and is in contact with laminin, while the "parenchymal" layer is characterised by the apical expression of osteopontin [15]. After the formation of a lumen, the nascent bile duct becomes symmetrical as "hepatoblasts" are replaced by "cholangiocytes", and the ductal structure matures along a cross-sectional axis and a cranio-caudal axis, extending from the hilum to the periphery. Thus, the progressive elongation of the duct requires a mechanism able to orient cell mitoses along the aforementioned axes in a coordinated manner. Through this mechanism, called "*planar cell polarity*" (PCP), the epithelial cells are uniformly oriented within the ductal plane to maintain the tubular architecture. PCP is a process that is controlled by the non-canonical Wnt pathway, and is defective in fibropolycystic liver diseases (see below) [16].

The mechanisms that regulate the termination of biliary development are not well known. Recent work from Kaestner's laboratory suggests that by inhibiting NF- κ B-dependent cytokine expression (specifically IL-6), the transcription factors Foxa1/2, may act as a termination signal in bile duct development. Mice with liver-specific deletion of both Foxa1 and Foxa2 showed an increased amount of dysmorphic bile ducts [17]. On the other

hand, a decrease in Notch signalling could change the fate of the non-duplicated ductal plate segments [18] and promote their differentiation towards alternative pathways.

Extrahepatic biliary tree

Cholangiocytes lining the extrahepatic bile ducts derive from the caudal part of the ventral foregut endoderm located between the liver and the pancreatic buds, a region that expresses a combination of transcription factors common to the pancreas and duodenum (Pdx-1, Prox-1, HNF-6). The extrahepatic part of the biliary tree develops before the intrahepatic part; the two systems merge at the level of the hepatic duct/hilum. Molecular mechanisms regulating the development of the extrahepatic bile ducts are less well known than those regulating the development of the intrahepatic bile ducts. Mice deficient in Pdx-1 [19] or Hes1 (a Notch-dependent transcription factor), HNF6, HNF-1 β , or Foxf1 (a transcription factor target for the sonic Hedgehog signalling) result in an altered development of the gallbladder and of the common bile duct [20–22].

Relationships between biliary and arterial development

Branches of the hepatic artery develop in close proximity to ductal plates. On one side, the biliary epithelium guides arterial development, on the other, the developing intrahepatic bile ducts are nourished by the peribiliary plexus (PBP), a network of capillaries emerging from the finest branches of the hepatic artery at the periphery of the liver lobule. PBP is crucial in maintaining the integrity and function of the biliary epithelium [23–25].

The patterning of the intrahepatic biliary tree develops in strict harmony with hepatic arteriogenesis. For example, inactivation of *Hnf6* or *Hnf1 β* , transcription factors involved in intrahepatic bile duct epithelium development, resulted in anomalies of hepatic artery branches paralleling bile duct abnormalities [1,12]. One of the signals linking ductal and arterial development in the liver is the vascular endothelial growth factor (VEGF), which cooperates with angiopoietin-1 (Ang-1). The developing bile ducts produce VEGF-A that in turn acts on endothelial cells and

their precursors, which both express the receptor VEGFR-2, to promote arterial and PBP vasculogenesis. At the same time, Ang-1 produced by hepatoblasts is likely to induce the maturation of hepatic artery terminations by recruiting mural pericytes to the nascent endothelial layer [12]. In ductal plate malformations (DPM), the dysmorphic bile ducts are surrounded by an increased number of vascular structures [5,26,27]. For example, in cystic cholangiopathies, the epithelium retains an immature phenotype typical of the embryonic ductal plate coupled with the production of VEGF and angiopoietins [26]. Production of angiogenic factor by immature cholangiocytes promotes an abundant pericytic vascularisation, thus providing the vascular supply around the growing liver cysts [27] (see below).

The anatomical and functional association between bile ductules and arterial vascularisation is maintained also in adult life and during liver repair. A malfunction of this system may cause ductopenia, as observed in chronic allograft rejection and in other chronic cholangiopathies caused by an ischaemic damage [28]. Ductular reaction is a common histopathological response to many forms of liver damage (see below); the increase in bile ductules at the portal tract interface is paralleled by an increased number of hepatic arterioles and capillaries [24,25]. These features are reproduced in an experimental rat model of selective cholangiocyte proliferation (α -naphthylisothiocyanate treatment), where an extensive neovascularisation of the arterial bed develops in strict conjunction with increased cholangiocyte mass [29]. Furthermore, expansion of the biliary tree after bile duct ligation in rat is also followed by substantial adaptive modifications of the PBP [25,30].

Transcription factors, growth factors and morphogens involved in the ontogenesis of the biliary epithelium

At the time of liver specification, i.e. when the endoderm becomes committed towards a liver cell fate, liver development is driven by several transcription factors including hepatocyte nuclear factor 1 β (HNF1 β) [21,31], Foxa1 and Foxa2 [17], and GATA-4 [32–34]. Extracellular signals, like fibroblast growth factor (FGF) secreted by the cardiogenic mesoderm [35], bone morphogenetic protein-4 (BMP-4) and extracellular matrix constituents [33,35], signal to competent cells and appear also to be involved in biliary differentiation.

The differentiation of the intrahepatic biliary epithelium and its tubular morphogenesis are finely regulated by signals exchanged between epithelial cells and a variety of other non-parenchymal cells. These signals encompass a series of morphogens and growth factors that regulate the *commitment of hepatoblasts* to the biliary lineage, or *tubule formation* by cholangiocytes, or more often, both processes. The portal mesenchyme generates a portal to parenchymal gradient of transforming growth factor- β (TGF- β 2 and TGF- β 3). TGF- β stimulates hepatoblasts to undergo a switch towards a biliary phenotype [36]. The developing ductal plates transiently express the TGF- β -receptor type II (T β RII); expression of T β RII is repressed in differentiated cholangiocytes.

HNF6 and HNF1 β are transcription factors able to regulate multiple steps of biliary development and morphogenesis. Their expression is upregulated in cholangiocytes and in progenitor cells committed to the cholangiocyte lineage. In mice, their absence is associated with cystic dysgenesis of the biliary tree

[21,37]. HNF6 and HNF1 β regulate distinct stages of bile duct morphogenesis. Whereas the absence of HNF6 caused an early defect in biliary cell differentiation, which can be somehow repaired, a defect in HNF1 β appears to be associated with a defect in the maturation of the primitive ductal structure [38]. Furthermore, HNF6 stimulates expression of *Pkhd1* while a deletion of HNF1 β causes an aberrant cystic development and defects in PCP, a feature associated with fibropolycystic liver diseases [16]. It is noteworthy that HNF1 β expression appears to be under the control of Notch signalling.

Notch signalling is a fundamental mechanism that confers cell fate instructions during the development of various tissues. Several studies using genetic mouse models and zebrafish demonstrated that Notch signalling is required at different stages during biliary tree development: from the formation of the ductal plate to ductal plate remodelling and tubule formation [39,40]. The Notch genes encode four transmembrane receptors (Notch 1, 2, 3, and 4), which can interact with a number of ligands (Jagged-1, Jagged-2, Delta-like 1, 3, and 4). Notch signalling requires the establishment of cell–cell contacts. Through this interaction among neighbouring cells, Notch receptors expressed by “receiving” cells are activated by the binding of ligands expressed on the surface of “transmitting” cells. Notch may stimulate cells to undergo a phenotypic switch through a process of “*lateral induction*”. Alternatively, Notch may promote the maintenance of the original phenotype through “*lateral inhibition*” [41]. In liver development, Notch signalling appears to control the ability of hepatoblasts and mature hepatocytes to differentiate into cholangiocytes by altering the expression of liver-enriched transcription factors, and to regulate the formation of biliary tubules [18]. Notch-dependent signalling mechanisms are summarised in Fig. 2. Notch drives the activation of Notch effector genes such as Hairy and Enhancer of Split homologues (*Hes1* and *Hey1*), via the transcription factor recombination signal binding protein for immunoglobulin kappa J (RBP-Jk), which in turn activates transcription factors specifically expressed by cholangiocytes, including HNF1 β and Sox9 [42].

Studies in mice have shown that Jagged-1 is expressed by periportal mesenchymal cells and interacts with Notch-2 expressed by hepatoblasts favouring their differentiation into ductal plate cells. Perturbations in Jagged-1/Notch-2 interactions cause Alagille syndrome (AGS), a genetic cholangiopathy characterised by ductopenia and defective peripheral branching of the biliary tree [43]. In fact, Jagged-1 inactivation in the portal vein mesenchymal cells, but not in the endothelial cells, results in a defective development of the bile ducts that do not mature beyond the initial formation of the ductal plate [44].

A major effect of Notch signalling is to modulate tubule formation, a property that is necessary to effectively repair the biliary tree. The effects of Notch receptors can be modified by other ligands such as the glycosyltransferases encoded by Fringe genes, and by the dosage of the *Jagged-1* gene [45]. The impact of Notch signalling on the intrahepatic bile duct branching is highlighted in a recent study by Sparks *et al.* [46], which demonstrated that the density of three-dimensional peripheral intrahepatic bile duct architecture during liver development depends on Notch gene dosage.

Canonical Wntless (Wnt)/ β -catenin participates in several stages of bile duct development. Specific Wnt ligands, such as Wnt3a, induce biliary differentiation, characterised by the appearance of K19 positivity and generation of duct-like

Review

Table 1. Primary cholangiopathies related to altered biliary development.

Group	Disease	Gene defect	Liver phenotype	Clinical manifestations
Ductopenic syndromes	AGS	<i>Jagged1</i> <i>Notch2</i>	Ductopenia, reduced HPC and RDC, increased IHBC, mild portal fibrosis	Pruritus Jaundice Failure to thrive
Malformative syndromes associated to DPM	Meckel syndrome	<i>MKS1</i>	Ductal plate remnants	Abort or perinatal death
	Joubert syndrome	<i>MKS3</i> and others	Ductal plate remnants	Abort or perinatal death
Polycystic liver diseases	ADPKD	<i>PKD1</i> <i>PKD2</i>	Liver cysts without peribiliary fibrosis	Cyst complications Mass effect Malnutrition Renal failure
	PLD	<i>PRKCSH</i> <i>SEC63</i>	Liver cysts without peribiliary fibrosis	Cyst complications Mass effect Malnutrition
Fibropolycystic liver diseases	ARPKD, CHF, CD	<i>PKHD1</i>	Biliary microhamartomas with peribiliary fibrosis	Acute cholangitis Intrahepatic lithiasis Portal hypertension Risk for evolution to CCA

AGS, Alagille syndrome; ADPKD, autosomal dominant polycystic kidney disease; PLD, polycystic liver disease; ARPKD, autosomal recessive polycystic kidney disease; CHF, congenital hepatic fibrosis; CD, Caroli's disease.

structures in mouse embryonic liver cell cultures. Wnt/ β -catenin appears to play a key role in biliary commitment, by repressing hepatocyte differentiation and promoting ductal plate remodelling [1,2]. The deletion of β -catenin in the developing hepatoblasts in transgenic mice leads to a paucity of bile ducts and to multiple defects in hepatoblast maturation, expansion, and survival [47]. Wnt can also signal through β -catenin-independent pathways. Non-canonical Wnt pathways seem to be crucial in the regulation of PCP [48], which is lost when *Pkhd1*, the gene encoding for fibrocystin (see below), is defective [16]. Notably, inversin, a cilium-associated protein regulating the left-right symmetry, modulates non-canonical Wnt signalling by interacting directly with Dvl [49] and if defective, causes aberrant development of intrahepatic bile ducts and cyst formation [16]. Canonical and non-canonical Wnt pathways are illustrated in Fig. 3.

Similar to Notch and Wnt, Hedgehog signalling (Hh) is a morphogenetic pathway involved in liver development. Signalling mechanisms of the Hh pathway are shown in Fig. 4. On the contrary to stromal and progenitor cells, mature epithelial cells do not retain the ability to respond to Hh signals. During development, sonic Hh is expressed in the ventral foregut endoderm, but its expression is then downregulated when the liver bud is formed [50]. In the foetal liver, Hh and Gli1, transcription factors, downstream of Hh signalling, are expressed in liver progenitor cells but their expression disappears as soon as the development proceeds. Therefore, activation of Hh signals is requested in a temporally restricted manner to promote the early hepatoblast proliferation, but then this pathway needs to be switched off to allow hepatoblasts to differentiate normally [50].

In addition to differentiation and tissue remodelling, biliary tubulogenesis requires cell proliferation. While the process of remodelling depends on complex interactions with mesenchymal and endothelial cells, tubule elongation is stimulated by a number of paracrine or autocrine factors, including oestrogens [51], insulin-like growth factor-1 (IGF1) [52,53], interleukin-6 (IL-6) [54], VEGF [26], which are able to stimulate cholangiocyte proliferation.

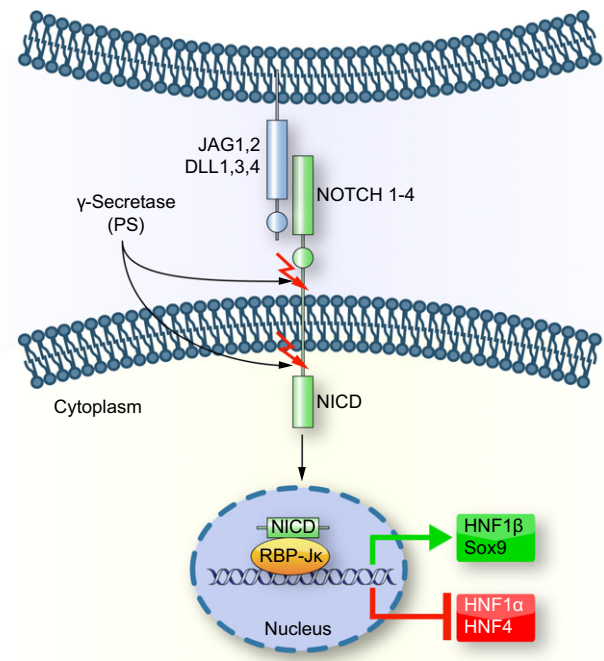


Fig. 2. Notch signalling. Jagged binding to a Notch receptor leads to the proteolytic processing and subsequent translocation into the nucleus of the Notch intracellular domain (NICD) of the receptor. Cleavage of NICD is an essential step in this process and is mediated by a γ -secretase enzyme in the cytoplasm. Once delivered into the nucleus, NICD forms a complex with its DNA-binding partner, the recombination signal binding protein for immunoglobulin kappa J (RBP-Jk). The formation of this complex leads to the upregulation of cholangiocyte-specific transcription factors, such as HNF1 β and the SRY-related HGM box transcription factor 9 (Sox9), and to the downregulation of hepatocyte-specific transcription factors such as HNF1 α and HNF4. Sox9 in particular, is the most specific and earliest marker of biliary cells in the developing liver, as it controls the timing and maturation of primitive ductal structures in tubulogenesis.

When tubule formation finishes, cholangiocytes become quiescent. Expression of ciliary proteins and control of cytokine secretion are important mechanisms involved in the termination

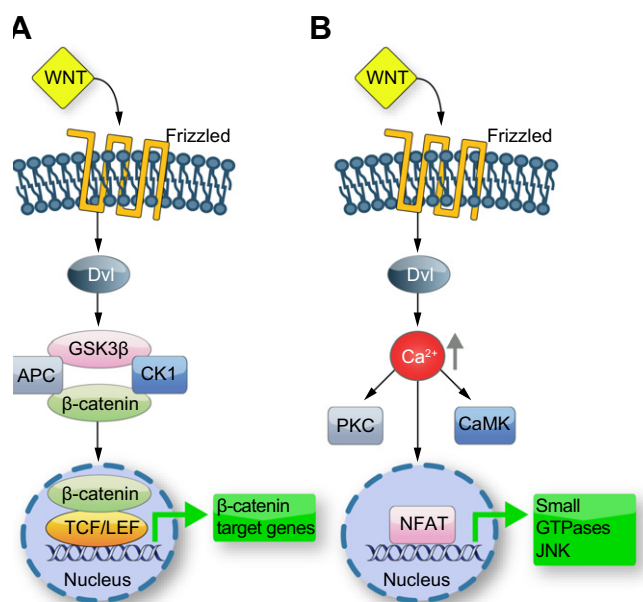


Fig. 3. Wnt signalling (canonical and non-canonical). Wnt signals in two different ways depending upon the activation (canonical) or the non-activation (non canonical) of β -catenin. In the canonical Wnt pathway (A), the binding of Wnt to Frizzled (Fzd) receptors activates Dishevelled (Dvl), which prevents the phosphorylation and the following ubiquitination of β -catenin. If β -catenin is not phosphorylated, it can thus accumulate in the cytoplasm and then translocate to the nucleus, where it activates Wnt target genes by interacting with the TCF/LEF family of transcription factors. In the non-canonical Wnt pathway (B), the binding of specific Wnt isoforms (Wnt 4, 5a, 11) to Fzd can activate Dvl but the downstream signal pathways involve small GTPases and the C-Jun N-terminal kinase (JNK) instead of β -catenin. Once activated, Dvl leads to increased intracellular Ca^{2+} levels that activate a number of proteins, including PKA, CaMK and NFAT. Acting as transcription factors, these proteins may activate downstream effectors that are crucial regulators of different cellular responses, such as planar cell polarity and cytoskeletal rearrangement.

of the developmental process. Foxa1 and Foxa2 are liver-specific transcription factors, which play a crucial role in establishing the developmental competence of the foregut endoderm and liver specification [17]. Interestingly, mice with liver-specific deletion of Foxa1/2 develop dysmorphic and dilated bile ducts surrounded by increased portal fibrosis. These changes are, in part, caused by the persistent expression of IL-6 by the biliary epithelium. In fact, the IL-6 promoter is negatively regulated by Foxa1/2 through binding to the glucocorticoid receptor. In absence of Foxa1/2, IL-6 expression is not suppressed leading to both autocrine effects on cholangiocytes (proliferation) and to paracrine effects on inflammatory cells and myofibroblasts (peribiliary fibrosis) [17]. This observation suggests that Foxa1/2 are among the transcription factors involved in the termination of bile duct development. Given the strong ability of reactive cholangiocytes to secrete large amounts of IL-6, this mechanism can be relevant also in liver repair (see below).

Ductular reaction, a reparative response to biliary and hepatocellular damage, is characterised by features reminiscent of biliary ontogenesis

Mechanisms of liver repair recapitulating liver developmental processes are an emerging concept. In fact, many molecular

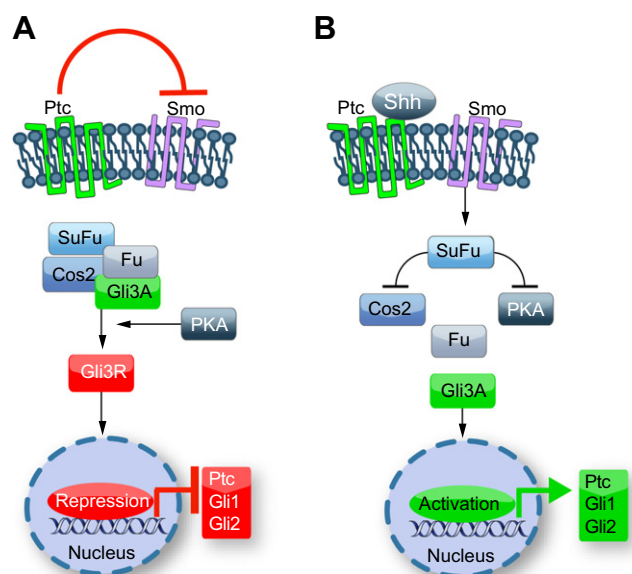


Fig. 4. Sonic Hedgehog (Shh) signalling. In physiological conditions (A), Patched (Ptc) receptor binds to and consequently suppresses the function of its co-receptor Smoothened (Smo). This maintains the bonding of the downstream regulator Glioblastoma-3 (Gli3) to a tetramer complex encompassing its suppressor factors, Suppressor Fused (Su(Fu)), Costal2 (Cos2), and Fused (Fu). In this complex, Gli3 is proteolytically cleaved by protein kinase A (PKA), and converted into the repressor form (Gli3R), which exerts an inhibitory effect on nuclear transcription factors regulating Hh-responsive genes (Ptc, Gli1, Gli2). When Shh interacts with Ptc (B), it prevents its inhibitory action on Smo. Following Smo activation, the tetramer complex is disassembled so that Su(Fu) inhibitory effect is restricted to Cos2 and PKA, thereby preventing the proteolytic cleavage of Gli3. Gli3 can thus enter the nucleus in the activated state (Gli3A) to promote activation of Hh target genes.

factors (growth factors, transcription factors, morphogens), transiently engaged in the development of the biliary epithelium during embryonic life, are reactivated in adulthood in response to acute or chronic liver damage [12,55].

In the adult liver, the division of mature epithelial cells, i.e. hepatocytes and/or cholangiocytes, drives normal tissue homeostasis and regeneration after acute and transient hepatocellular or biliary damage. On the other hand, in chronic liver diseases, liver repair relies on the activation of *hepatic progenitor cells* (HPC). HPC are bipotent cells located in close proximity to the terminal cholangioles at their interface with the canals of Hering. HPC are able to amplify and differentiate into cells committed to hepatocellular or biliary lineages [56–59]. In humans, differentiation towards hepatocytes occurs via *intermediate hepato-biliary cells* (IHBC), whereas differentiation towards the biliary lineage leads to the formation of *reactive ductules* (RDC) [59]. These cellular elements encompass the “*hepatic reparative complex*” and can be recognised from their expression of cytokeratin-7 (K7) and/or -19 (K19), cytoskeletal proteins present in the biliary lineage, but not in hepatocytes (Fig. 5) [9,55]. Although transdifferentiation from hepatocytes (formerly recognised as “*ductular metaplasia*” of hepatocytes) may occur under certain circumstances, reactive ductules are believed to derive mostly from the progenitor cell compartment.

Initially, RDC organise in clusters and do not encircle a lumen. As RDC's participation in tissue remodelling develops, they eventually reorganise into a richly anastomosing tubular network.

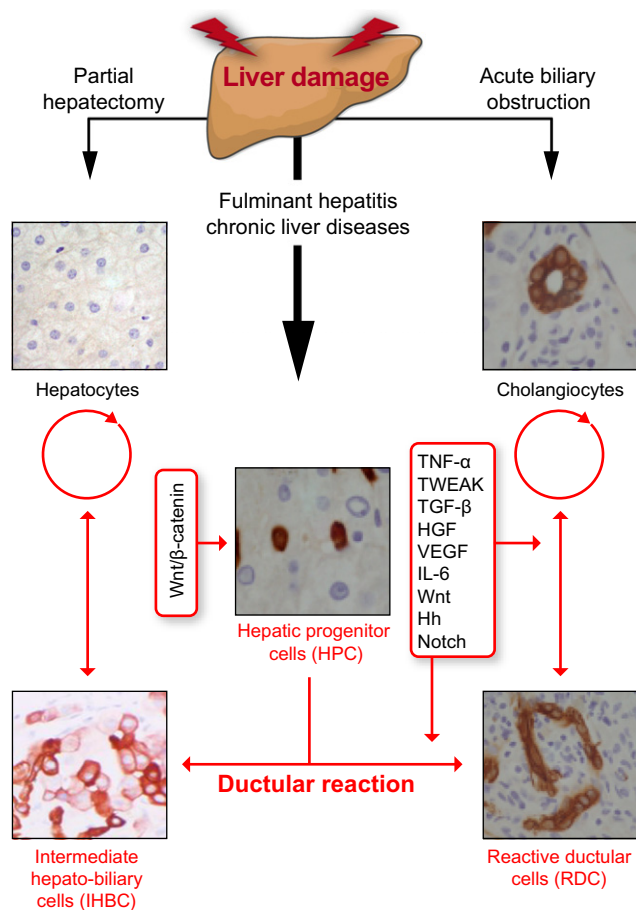


Fig. 5. Epithelial phenotypes involved in liver repair driven by the activation of hepatic progenitor cells ("Hepatic Reparative Complex"). In acute and chronic liver diseases, especially in fulminant hepatic failure, liver repair is driven by the activation of hepatic progenitor cells (HPC) to differentiate into hepatocytes and/or cholangiocytes. However, hepatocyte and cholangiocyte proliferation can be directly stimulated without exploiting HPC activation in experimental models, such as partial hepatectomy and acute biliary obstruction, respectively. HPC are small epithelial cells with an oval nucleus and scant cytoplasm, similar to oval cells in rodents treated with carcinogens. HPC originate from a niche located in the smaller branches of the biliary tree and in the canals of Hering. HPC behave as a bipotent, transit amplifying compartment. Differentiation of HPC towards hepatocytes occurs via intermediate hepatobiliary cells (IHBC), while differentiation towards the biliary lineage leads to the formation of reactive ductular cells (RDC). HPC, IHBC and RDC constitute the "hepatic reparative complex", and can be distinguished by morphology and pattern of K7 expression. Whereas Wnt signalling is a key regulator of proliferation of HPC, Notch and Hh signalling are mostly involved in biliary differentiation through RDC generation, along with other cytokines released from the inflammatory microenvironment (TNF- α , TWEAK, TGF- β , HGF, VEGF, IL-6) (see text for details).

Tubule formation during biliary repair is a fundamental process aimed at generating a compensatory increase of the ductal mass to prevent the development of extensive liver necrosis due to the leakage of bile into the parenchyma. It is important to recognise that activation and proliferation of HPC is not sufficient to repair biliary damage unless progenitor cells and reactive cholangiocytes acquire the ability to form new branching tubular structures, thus restoring the ductal mass [60].

In addition to replacement of damaged cells and development of branching tubules, liver repair requires the generation of a fibro-vascular stroma able to sustain and feed the remodelling

ductal structures. As a consequence, RDC *de novo* express a variety of cytokines, chemokines, growth factors, angiogenic factors and adhesion molecules, along with their receptors. This property enables RDC to establish an extensive crosstalk with other liver cell types, including hepatocytes, stellate cells and endothelial cells (Fig. 6) [61]. When the biliary tree is damaged, these interactions become functionally relevant leading to a significant expansion of the reactive cholangiocyte compartment. Most of the molecular signatures expressed by reactive cholangiocytes, including VEGF [12], TGF- β [62], connective tissue growth factor (CTGF) [63], stromal cell-derived factor 1 (SDF-1) [64], IL-6 [65], tumour necrosis factor- α (TNF α) [65], neural cell adhesion molecule (NCAM) and Bcl-2 [55], are transiently expressed by ductal plate cells in foetal life, consistent with the concept that ductular reaction recapitulates liver ontogenesis [55,66–68] (Fig. 7).

The molecular mechanisms that activate ductular reaction require a finely coordinated process that also shares a number of similarities with biliary embryogenesis. Ductular reaction is elicited by inflammatory signals released from the local microenvironment. TNF- α [69], TWEAK [70], TGF- β [71], HGF [71], VEGF [26,72], sonic Hedgehog (shh) [73], and Wnt/β-catenin [74] signalling are among the key signalling pathways. As mentioned above, proliferation of reactive cholangiocytes may be achieved by switching off transcription factors (Foxa1/2) used to terminate development. Unfortunately, activation/deactivation of developmental mechanisms in the context of non-resolving inflammation, leads to the development of portal fibrosis in cholangiopathies. The ability of "reactive" cholangiocytes to recruit inflammatory, vascular and mesenchymal cells, with which they exchange a variety of paracrine signals, subsequently leads to excessive collagen deposition and stimulation of angiogenesis, and ultimately to cirrhosis [75].

Several morphogens involved in biliary development, such as Wnt, Hedgehog, and Notch, also play a pivotal role in liver repair. Recent findings suggest that the Wnt/β-catenin pathway is strongly involved in the normal activation and proliferation of adult HPC in both acute and chronic human liver diseases [76]. Oval cell activation in β-catenin conditional knockout mice was dramatically reduced, after chemical treatment to induce acute liver damage [77].

Whereas Wnt signalling is a key regulator of proliferation of HPC, Notch signalling is mostly involved in biliary differentiation. The role of Notch is underscored by the peculiar liver phenotype observed in AGS. AGS is characterised by a marked reduction in RDC and HPC, in sharp contrast with biliary atresia, which is a congenital cholangiopathy with similar levels of homeostasis but much faster evolution to biliary cirrhosis [43]. This difference is likely related to a Notch-dependent block in cell fate determination upstream of HNF1β. Recent data from our group show that following treatment with cholestatic agents, HPC activation and tubule formation are dramatically impaired in mice with a liver-specific defect in RBP-Jk [78].

In addition to Notch, Hh signalling is also relevant in congenital cholangiopathies. Perturbations in the Hh signalling have been associated with DPM in Meckel syndrome (MKS), a rare autosomal recessive disease caused by a defect in *MKS1* gene encoding a protein associated with the base of cilia. MKS causes perinatal lethality and is characterised by a complex syndrome including polycystic kidneys, occipital meningoencephalocele, postaxial polydactyly in addition to DPM [79]. In liver repair, activation of the Hh signalling promotes the expansion of a subset of

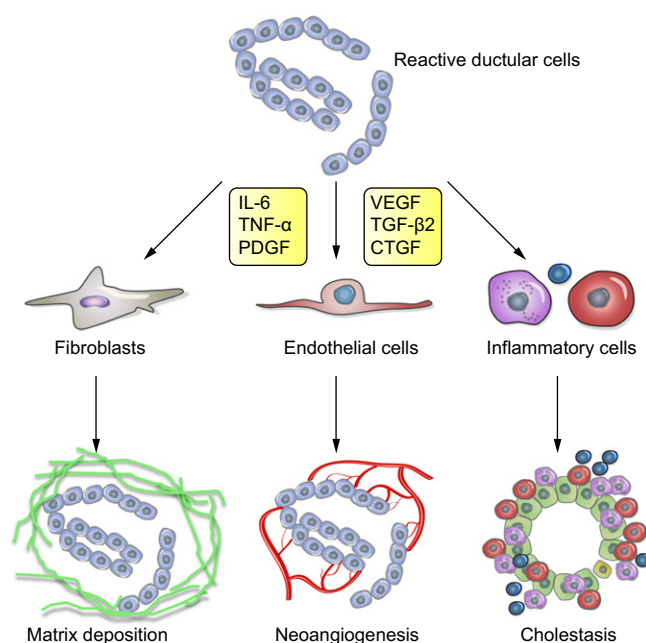


Fig. 6. Reactive ductular cells acquire the ability to exchange a range of paracrine signals with mesenchymal, vascular and inflammatory cells. Owing to the *de novo* expression of a variety of cytokines, chemokines, growth factors, angiogenic factors, together with a rich expression of many of the respective cognate receptors, reactive ductular cells can establish an extensive crosstalk with other liver cell types, particularly with stellate cells, endothelial cells and inflammatory cells. In response to biliary damage, these interactions become functionally relevant leading to the generation of a fibro-vascular stroma able to sustain and feed the ductular reaction, and to the recruitment of a peribiliary inflammatory infiltrate which further enhances the bile duct damage.

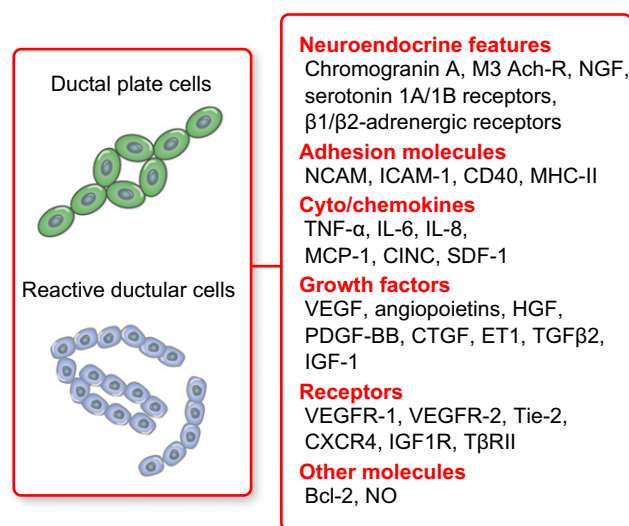


Fig. 7. Phenotypic changes of ductular reactive cells shared with ductal plate cells. An important feature of reactive cholangiocytes is the foetal reminiscence of their phenotype. Reactive ductular cells express neuroendocrine features, adhesion molecules, cytokines and chemokines, receptors and other metabolically active molecules, which are transiently expressed by ductal plate cells during embryonic development.

immature ductular cells that co-express mesenchymal markers and may be profibrogenic [80]. This mechanism is relevant in

biliary atresia, where excessive activation of Hh signalling halts bile duct morphogenesis and promotes accumulation of immature ductular cells with a mesenchymal phenotype, which in turn enhance fibrogenesis [81]. These data suggest that aberrant activation of Hh signalling may be responsible for biliary fibrosis featuring developmental cholangiopathies.

Primary cholangiopathies related to an altered development of the biliary epithelium

Cholangiopathies related to altered biliary development [5,82] (Table 1) represent important disease models; understanding their pathogenetic mechanisms offers important clues on how specific genes regulating developmental processes are involved in biliary repair and reaction to damage. Recently, based on the phenotype of several mouse models with specific defects of biliary morphogenesis, DPM have been classified into three groups, with: (a) defective differentiation of biliary precursors cells (HNF6 deficiency), (b) defective maturation of primitive ductal structure (HNF1 β deficiency) or (c) defective duct expansion during development with preserved biliary differentiation (cystin-1 deficiency) [38]. Most DPM are embryonically lethal or are part of complex syndromic diseases. Herein, we will outline the most recent advances in the pathogenesis of congenital cholangiopathies that may be clinically relevant for both paediatric and adult clinical hepatologists. The most common DPM are the Von Meyenburg complexes, also known as biliary microhamartomas (Fig. 8B). These are benign lesions characterised by irregularly shaped and dilated biliary structures, embedded in a dense fibrous stroma. Their distribution is generally focal, but when diffuse they can be associated with cystic lesions, as seen in congenital hepatic fibrosis (see below) [6]. Von Meyenburg complexes have also been found in higher numbers in the liver of patients with autosomal dominant polycystic kidney disease [83,84]. The association of Von Meyenburg complexes to cholangiocarcinoma is uncommon, but it has been sporadically described [85–87].

Autosomal recessive polycystic kidney disease (ARPKD), congenital hepatic fibrosis (CHF) and Caroli's disease (CD)

The presence of ductal plate remnants, Von Meyenburg complexes, and biliary cysts variably associated with an intense peribiliary fibrosis, are the main features of fibropolycystic diseases [6]. These include ARPKD, and its hepatic variants, CHF and CD, and a variety of congenital syndromes (Meckel and Joubert syndromes), which are often embryonically lethal. In CHF and CD, biliary malformations are associated with progressive portal fibrosis leading to portal hypertension, without progressing to frank cirrhosis. In patients with CD, the risk of developing cholangiocarcinoma is substantially increased (about 10%).

CHF and CD are caused by mutations in *PKHD1*, a gene encoding for fibrocystin (FPC). FPC is a large membrane, receptor-like protein expressed by the basal body of cilia, subcellular organelles that sense the direction of ductal bile flow, and by centromeres of renal tubular and bile duct epithelial cells [88–90]. Although FPC functions are largely unknown, some of its properties have been recently outlined [91]. FPC is thought to be involved in a variety of functions, from proliferation to secretion, terminal differentiation, tubulogenesis, and interactions with the extracellular matrix. Silencing *Pkhd1* in cultured mouse renal

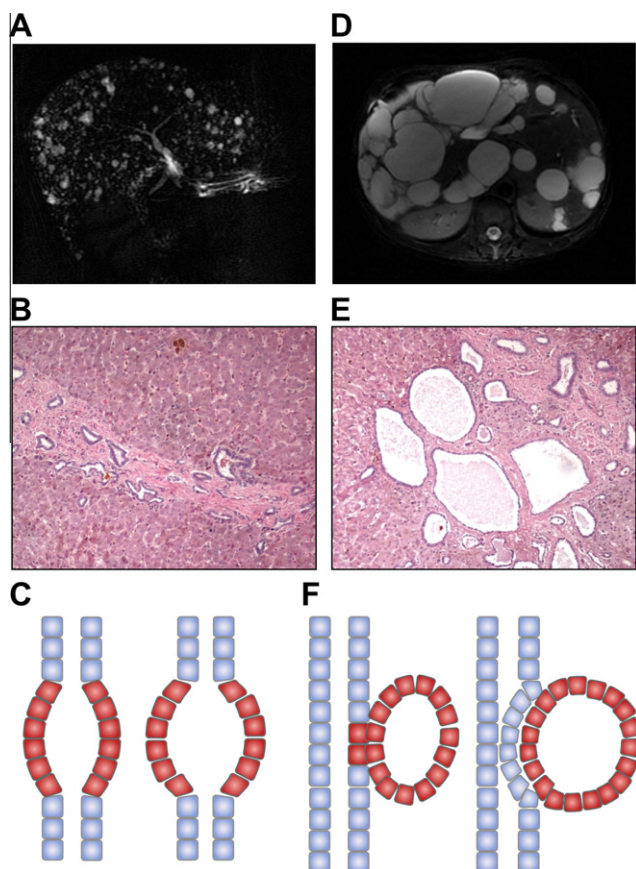


Fig. 8. Main differences in liver phenotype between ARPKD and ADPKD. On magnetic resonance imaging, liver cysts appear as focal lesions with regular margins, which are of small size and in continuity with the biliary tree in ARPKD (A), whereas they are large, of different size and scattered throughout the hepatic parenchyma leading to extensive cyst substitution in ADPKD (D). At histological examination, irregularly shaped biliary structures (microhamartomas) surrounded by an extensive deposition of fibrotic tissue, are present in the portal tracts in ARPKD (B, H&E; magnification: 100 $\times$), while in ADPKD, biliary cysts appear as large, circular biliary structures, lined by cuboidal or flattened epithelium, with negligible amount of peribiliary fibrosis (E, H&E, magnification: 100 $\times$). The pathogenetic mechanism underlying cyst formation is characterised by progressive segmental dilation of biliary structures which maintain their connection to the biliary tree in ARPKD (C), whereas biliary cysts detach from the bile duct and then progressively increase in size in ADPKD (F). Alternatively, based on previous histopathological studies [110–112], liver cysts in ADPKD may also derive from dilatation of components of Von Meyenburg complexes.

tubular cells altered cytoskeletal organisation and impaired cell–cell and cell–matrix contacts [92]. Recent evidences indicate that FPC is also involved in planar cell polarity. In the *Pck* rat, a model orthologue of ARPKD, FPC deficiency is strongly correlated with the loss of PCP at the renal level. This leads to a perturbed mitotic alignment on circumferential tubular cell number expansion, which is ultimately responsible for renal tubular enlargement and cyst formation [16]. The mechanistic relationships between biliary dysgenesis and portal fibrosis are not understood and this is an area of current investigation.

Autosomal dominant polycystic kidney disease (ADPKD) and polycystic liver disease (PLD)

Autosomal dominant polycystic kidney disease (ADPKD) and polycystic liver disease (PLD) are inherited with an autosomal

dominant transmission. These conditions do not cause significant liver fibrosis, but rather an enormous increase in liver mass. The formation and progressive enlargement of multiple cysts scattered throughout the liver parenchyma characterise both ADPKD and PLD [93]. Despite extensive cyst substitution of the hepatic parenchyma, liver function is generally well preserved and portal hypertension is rare.

The patients are usually asymptomatic, unless acute and chronic complications (including cyst infections or bleeding and mass effect) develop. ADPKD is caused by mutations in the *PKD1* or *PKD2* genes [94,95]. The transcribed products of these two genes, polycystin-1 (PC1) and polycystin-2 (PC2), are membrane proteins located in the renal tubular and biliary epithelia. PC1 and PC2 regulate signalling pathways that are involved in epithelial cell morphogenesis, differentiation and proliferation. In conditional, liver specific mice defective for PC-1 or PC-2, polycystic liver disease develops even if PC1 and PC2 are deleted after birth, indicating that PC expression maintain a fundamental morphogenetic role also during adult life [27,96]. Thus, altered PC function may cause a lack of differentiating signals favouring the maintenance of an immature and proliferative phenotype by biliary epithelial cells ultimately responsible for cyst formation. In fact, cholangiocytes lining the liver cysts present strong phenotypic and functional similarities with ductal plate/reactive ductules [26]. Among them are an aberrant secretion of several cytokines and chemokines, including IL-6 [97], IL-8 [97], and CXCR2 [98], a marked overexpression of oestrogen receptors, IGF1, the IGF1 and growth hormone receptors as well as VEGF and its cognate receptor, VEGFR-2 [26]. In particular, VEGF potentially stimulates the progression of liver cysts in ADPKD via autocrine stimulation of cholangiocyte proliferation and paracrine promotion of pericystic angiogenesis. We have shown that in cystic cholangiocytes from *Pkd2*-defective mice a MEK/ERK1/2/mTOR pathway is overactive and is responsible for increased hypoxia-inducible factor 1 α -dependent VEGF production and increased VEGFR-2-mediated autocrine stimulation of cyst growth. This mechanism represents a potential target for therapy, as the blockade of angiogenic signalling using a competitive inhibitor of VEGFR-2, or the administration of mTOR inhibitors block the growth of liver cysts in *Pkd2*KO mice, and reduces the proliferative activity of the cystic epithelium [96]. Overall, these data indicate that polycystic liver diseases should be considered as congenital diseases of cholangiocyte signalling [99]. Phenotypic changes in cyst cholangiocytes with functional relevance for cyst formation and progression are summarised in Table 2. Each pathway is also therapeutically relevant as a potential target amenable of pharmacological interference aimed at reducing disease progression. Future studies will clarify whether the use of agents interfering with VEGF or mTOR signalling can be clinically useful and if their use can be extended to other cholangiopathies [27].

In PLD, genetic defects do not involve ciliary proteins. PLD is caused by mutations in *PRKCSH*, a gene encoding for protein kinase C substrate 80K-H also called hepatocystin [100], or in the *SEC63* gene [101]. SEC63 encodes a component of the molecular machinery regulating translocation and folding of newly synthesised membrane glycoproteins. Hepatocystin and SEC63 are expressed on the endoplasmic reticulum (ER) [102], and defects in other enzymatic activities associated with the ER, such as xylosyl transferase 2 responsible for initiating heparin sulphate and chondroitin sulphate biosynthesis, have been linked

Table 2. Phenotypic changes in cyst cholangiocytes.

- Less differentiated phenotype
- VEGF and VEGFR2 expression
- Cytokines and chemokines secretion
- Lower cytoplasmic $[Ca^{2+}]$
- Increased cAMP levels
- Increased expression of mTOR, pERK1/2
- Increased proliferation/apoptosis
- Changes in ER functions
- Altered non-canonical Wnt signaling

to the development of renal and liver cysts [103]. PLD serves to highlight the important concept that cystic liver disease does not necessarily derive from ciliary dysfunction, but that the defective proteins are expressed in multiple cellular locations, including the ER.

Alagille syndrome (AGS)

Alagille syndrome (AGS) is a complex, multisystemic disorder inherited as an autosomal dominant trait and characterised by ductopenia. AGS exhibits a wide range of extrahepatic manifestations (cardiac, vascular, skeletal, ocular, facial, renal, central nervous system), hence the term of “syndromic bile duct paucity” [104,105]. The hepatic phenotype is recognised by variable degrees of cholestasis, jaundice and pruritus. Fibrosis is not a prominent feature of AGS and frank evolution to cirrhosis is rare, however, rare cases lead to liver transplantation [106]. In nearly 80% of AGS patients, a mutation in the genes *JAGGED1* [107,108], or less frequently, *NOTCH2* can be identified [109]. AGS differs from other cholestatic cholangiopathies in the extent and severity of ductular reaction. The near absence of HPC and RDC in AGS (Fig. 5) is conversely coupled with an extensive accumulation of IHBC. IHBC do not express the biliary specific transcription factor HNF1 β , whose expression is controlled by Notch signalling [43]. This imbalance in the cellular elements of the “hepatic reparative complex”, which is a peculiar histopathological feature of AGS, supports the concept that Notch signalling plays an essential role in liver repair by regulating the generation of biliary committed precursors as well as the branching tubularisation [78].

Conclusions

Several cholangiopathies are caused by a malfunction of developmental mechanisms. In these diseases, cholangiocyte dysfunction is often caused by a specific genetic defect relevant to morphogenesis. Also, cholangiocytes express a range of phenotypic features reminiscent of foetal behaviour. In different forms of acquired liver damage, liver repair exploits several developmental mechanisms, in a sort of “recapitulation of ontogenesis”. Ductular reactive cells generated in response to liver damage express several of the autocrine and paracrine signals transiently expressed by ductal plate cells during liver development. Therefore, understanding the molecular mechanisms regulating development of the biliary tree may help solve several challenging

issues facing modern Hepatology. These may range from the search of novel treatments for patients with biliary diseases, to the need of limit/prevent pathologic repair in chronic liver diseases, to the design of bioartificial liver support devices, to strategies for liver regenerative medicine.

Key Points

- Intrahepatic and extrahepatic bile ducts have different embryologic origins. Whereas cholangiocytes lining the intrahepatic bile ducts derive from hepatoblasts, extrahepatic cholangiocytes derive from the caudal part of the ventral foregut endoderm
- Signals from the portal mesenchyma stimulate hepatoblasts to differentiate into ductal plate cells. After ductal plate duplication, biliary tubule formation requires a process of transient asymmetry that proceeds from the hilum to the periphery of the liver and is completed only after birth. A number of transcription factors and signaling mechanisms have been identified. Notch signaling is involved in multiple steps, from ductal plate formation to ductal plate remodeling and tubularisation
- Ductal plate malformations are caused by genetic defects in proteins expressed in multiple cellular locations, including primary cilia and the endoplasmic reticulum. Phenotypic features reminiscent of ductal plate cells are also expressed by cholangiocytes layering liver cysts and microhamartomas in developmental cholangiopathies. Cholangiopathies are a group of diseases in which cholangiocyte dysfunction is often induced by a specific genetic defect relevant to morphogenesis
- Ductular reaction is a common compensatory and reparative response to different forms of liver damage, which recapitulates ontogenesis. In fact, ductal plates transiently express several of the autocrine and paracrine signals generated by ductular reactive cells in response to liver damage, during liver development
- Understanding the molecular mechanisms regulating development of the biliary tree in embryonic life provides invaluable clues on the mechanisms underlying epithelial reaction and liver damage in adulthood

Conflict of interest

The authors declared that they do not have anything to disclose regarding funding or conflict of interest with respect to this manuscript.

Review

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